

VIKRAM RAVI

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SUMMARY

Applied AI Engineer with 7+ years of production engineering experience specializing in Agentic RAG systems, LLM application development, LangGraph multi-agent orchestration, and building AI-powered automation workflows in regulated enterprise environments. Engineered enterprise-grade Agentic RAG platforms utilizing LangGraph for multi-agent orchestration, implementing RAGAS-based quality gates, LangSmith observability, and SSE streaming to ensure production-ready performance. Builds and evaluates end-to-end AI systems using Claude and Gemini APIs on a strong foundation in Python, FastAPI, React, and TypeScript.

TECHNICAL SKILLS

AI & Data Engineering: Claude API, Google Gemini API, LangChain, LangGraph, LangSmith, Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, Tool Use / Function Calling, Hybrid Retrieval (BM25 + Vector Search), Vector Embeddings, Semantic Search, MMR Re-Ranking, RAGAS, LLM Evaluation (LLM-as-a-Judge), Prompt Engineering, NLP, Model Context Protocol (MCP), PII Redaction, Pydantic, Redis Caching

Frontend & UI: React, Next.js, Redux, Vite, GraphQL (Apollo Client), Tailwind CSS, JavaScript, TypeScript, HTML5, CSS3, Storybook

Backend & API: Python, FastAPI, Node.js, Express.js, REST APIs, GraphQL, SQLAlchemy, Postman

Cloud & Infrastructure: AWS (EC2, S3), Docker, Kubernetes, GitHub Actions, Jenkins, Git, GitHub, Claude Code, Cursor, Sentry

PROFESSIONAL EXPERIENCE

Scotiabank — Senior Full Stack Developer | Applied AI

Aug 2021 – Present

- Contributed to an enterprise-grade Agentic RAG platform — architected a LangGraph-based multi-agent pipeline featuring planner nodes, retrieval agents, and iterative answer-refinement loops to support complex, multi-tenant deployment scenarios.
- Evaluated and implemented hybrid retrieval (BM25 + vector search) with MMR re-ranking to optimize retrieval grounding and noise reduction across large-scale, heterogeneous document repositories.
- Designed a two-level Redis caching layer and SSE streaming pipeline to optimize latency across high-volume RAG queries.
- Engineered an asynchronous guardrails layer with PII scrubbing and prompt-injection interception before LLM invocation; evaluated retrieval quality offline with RAGAS metrics and LLM-as-a-judge scoring to assess agent quality.
- Designed human escalation and review patterns for AI-assisted workflows, ensuring outputs remained auditable and overridable by subject-matter experts.
- Integrated LangSmith tracing across the LangGraph pipeline to monitor latency, token usage, and retrieval quality, enabling end-to-end production observability.
- Architected a frontend application from the ground up — module structure, component hierarchy, routing, and coding standards.
- Applied Clean Architecture principles and designed custom React hooks to separate business logic from UI, reducing cross-module coupling.
- Integrated the frontend with GraphQL APIs via Apollo Client and partnered with backend engineers to refine GraphQL contracts.
- Containerized AI services with Docker and Kubernetes, establishing deployment best practices and reproducible build pipelines integrated with GitHub Actions.
- Delivered AI-powered solutions within a highly regulated fintech environment, navigating compliance constraints and enterprise security standards.
- Optimized web performance across 10+ high-traffic pages using Lighthouse and Web Vitals, achieving up to 35% reduction in LCP and TTI.
- Enforced clean architecture principles by decoupling presentation layers from business logic, resulting in enhanced scalability.

Ionixx Technologies — Programmer

Jun 2017 – Apr 2019

- Participated in Agile development cycles, contributing to all phases of the SDLC including analysis, development, testing, and deployment.
- Drove the migration of a legacy locator system to Next.js, achieving a 30% improvement in page load times.
- Optimized Node.js backend performance with Redis caching, reducing database query times by ~40%.
- Designed and delivered Express.js middleware services and a real-time Project Locator UI with interactive map integration.
- Engineered responsive, cross-browser UI components using React.js, Bootstrap, and SASS/LESS.

PROJECTS

Deep Research AI Agent | Python · LangGraph · Gemini 2.0 · Gradio · Tavily

- Designed a 4-stage autonomous research pipeline that executes multi-step web scraping and synthesizes structured, cited briefs.
- Engineered a critic-driven rework loop using LangGraph state graphs and async parallel execution, optimizing token throughput.
- Implemented an automated LLM-as-a-judge evaluation framework scoring generated briefs on relevance and citation quality.

AI Ticket Classifier | Python · FastAPI · LangGraph · Anthropic Claude API · React

- Designed a governed LangGraph pipeline that classifies support tickets by category, priority, and routing team with sentiment analysis.
- Added a safety layer performing PII redaction and prompt-injection detection prior to classification, validated via Pydantic.
- Exposed a FastAPI service with SQLAlchemy persistence for classification history and a React 19 frontend.
- Implemented LLM-as-a-judge evaluation scoring classifier outputs on faithfulness and label accuracy.

EDUCATION

Master of Engineering, Industrial Engineering — University of Windsor

Apr 2019 – Aug 2020

Bachelor of Engineering, Mechanical Engineering — Anna University

Jun 2013 – May 2017

CERTIFICATIONS

- AWS Certified Developer – Associate
- Building with the Claude API — Anthropic Academy (2026)
- Model Context Protocol (MCP) — Anthropic Academy